

Unit 27 — Multiplicative Relations | Resource Appendix | Stage 3 Year B | Word Labs SOLO Tracker

Videos (full URLs in the SOLO Tracker) and Corbettmaths worksheets (WS) listed in outcome sequence. Worksheets follow below, each with a divider page.

Band	Code	Outcome	Type	Resource Label / Short URL
Red	R1	Area model and algorithms	Video	Box Method Multiplication 2-Digits x 2-Digits Math with Mr. J youtu.be/n3q3XzzIGSY
Red	R1		Video	2-Digit by 2-Digit Multiplication Math with Mr. J youtu.be/PZjIT9CH6bM
Red	R1		WS	Corbettmaths — Multiplication 1 corbettmaths.com/.../multiplication-1-pdf.pdf
Red	R3	Factors, products, primes	Video	Mini Maths: Year 6 — Prime, Composite and Square Numbers youtu.be/b-XavTWnTMY
Red	R3		Video	Prime and Composite Numbers youtu.be/44jD-K2leEM
Red	R3		WS	Corbettmaths — Multiples, Factors and Primes (Exam Style) corbettmaths.com/.../multiples-factors-primes-pdf.pdf
Red	R4	Patterns with multiples	Video	Year 5: Recognising Factors, Multiples and Prime Numbers youtu.be/3KxEwpX6Ags
Red	R4		WS	Corbettmaths — Factors, Multiples and Primes corbettmaths.com/.../factors-multiples-primes.pdf
Red	R5	Multiply up to 4 digits	Video	Box Method Multiplication 3-Digits x 2-Digits Math with Mr. J youtu.be/zfYG9IzMMuO
Red	R5		Video	2-Digit Multiplication: Step-By-Step Review Math with Mr. J youtu.be/yIZ3r9T_OJs
Red	R5		WS	Corbettmaths — Multiplication 2 corbettmaths.com/.../multiplication-2-pdf.pdf
Yel	Y1	Division strategies and word problems	Video	Division Strategies Multiplication and Division Year 5 youtu.be/aTtuFEZVKaM
Yel	Y1		Video	Learn 5th Grade Math — Long Division Word Problem youtu.be/xD5GI-mWRnk
Yel	Y1		WS	Corbettmaths — Division corbettmaths.com/.../division-pdf.pdf
Yel	Y2	Remainders in division	Video	Interpreting Remainders in Division Word Problems youtu.be/0dPQEKJTFjg
Yel	Y2		Video	5th Grade Math 2.7 — Interpret the Remainder in Division youtu.be/FkRmFMwYcTA
Yel	Y2		WS	Corbettmaths — Division Remainders corbettmaths.com/.../Division-Remainders-pdf.pdf
Yel	Y3	Order of operations	Video	Order of Operations (BODMAS) Year 5 and 6 Home Learning youtu.be/VY-Da9ZmFFs
Yel	Y3		Video	Order of Operations PEMDAS 5th Grade Math youtu.be/OdFXImly2-I
Yel	Y3		WS	Corbettmaths — Order of Operations (BODMAS) corbettmaths.com/.../order-of-operations-pdf.pdf
Yel	Y3		WS	Corbettmaths — Order of Operations Exercise corbettmaths.com/.../Order-of-Operations-Exercise-211-pdf.pdf
Yel	Y4	Inverse operations	Video	Inverse Operations — Multiplication and Division The Maths Guy youtu.be/RsR8zwWj_Bw
Yel	Y4		Video	Solve Missing Number Problems for Multiplication and Division youtu.be/M5upWuCP4KA
Yel	Y4		WS	Corbettmaths — Division Exam Style Questions corbettmaths.com/.../division-pdf1.pdf
Yel	Y5	Rates word problems	Video	How to Multiply and Divide Whole Numbers Math with Mr. J youtu.be/gmo5BTdBbmE
Yel	Y5		Video	Multiplication Strategies — 6 Methods youtu.be/Ram-TykPgl8
Yel	Y6	Division as fractions	Video	Fractions as Division MightyOwl Math 5th Grade youtu.be/gXotbpz5D0M
Yel	Y6		Video	Numerator and Denominator of a Fraction Khan Academy youtu.be/3XOt1fjWKi8

Band	Code	Outcome	Type	Resource Label / Short URL
Grn	G1	Multiply and divide integers	Video	Multiplying Negative Numbers — Corbettmaths youtu.be/ElAvJfpGD0w
Grn	G1		Video	Directed Numbers Multiplication and Division youtu.be/NUqAsCgdArM
Grn	G1		Video	Year 7 Directed Numbers — All Four Operations youtu.be/4XC8QqgGWJ0
Grn	G1		WS	Corbettmaths — Negative Numbers: Mult and Division corbettmaths.com/.../Negatives-multiplication-and-division-pdf.pdf
Grn	G2	Order of operations with integers	Video	Order of Operations using BODMAS youtu.be/-ppkSPdMs5M
Grn	G2		Video	How To Use BODMAS for Order of Operations youtu.be/nHXT8am9X0o
Grn	G2		WS	Corbettmaths — Order of Operations (BODMAS) corbettmaths.com/.../order-of-operations-pdf.pdf
Grn	G2		WS	Corbettmaths — Arithmetic with Negatives (Exam Questions) corbettmaths.com/.../arithmetic-with-negatives-pdf.pdf
Grn	G3	Pronumerals and expressions	Video	GRADE 7 MATH — Introduction to Algebra: Algebraic Expressions youtu.be/A7cB5iglREI
Grn	G3		Video	Adding and Subtracting Algebraic Expressions (Pronumerals) youtu.be/0keMdSQvAKQ
Grn	G3		Video	Forming Algebraic Expressions — Corbettmaths youtu.be/Zlv-ZCRzS80
Grn	G3		WS	Corbettmaths — Substitution corbettmaths.com/.../substitution-pdf2.pdf
Grn	G4	Evaluate by substitution	Video	Substitution into Expressions (part 2) — Corbettmaths youtu.be/28DkE4vMN6o
Grn	G4		Video	Substitution into Expressions — Corbettmaths youtu.be/ZkC2FX5TOJ8
Grn	G4		WS	Corbettmaths — Substitution corbettmaths.com/.../substitution-pdf2.pdf
Grn	G5	Simplify: like terms	Video	Combining Like Terms Math with Mr. J youtu.be/1MmhAq-XVN0
Grn	G5		Video	Collecting Like Terms — Corbettmaths youtu.be/zxJNJMDj2Ec
Grn	G5		Video	Year 7 Maths — Collecting Like Terms youtu.be/dxn2hdxejAo
Grn	G5		WS	Corbettmaths — Collecting Like Terms corbettmaths.com/.../collecting-like-terms-pdf3.pdf
Grn	G6	Distributive law and brackets	Video	C1 Simplifying Expressions — Corbettmaths youtu.be/0Nc22d2mdm8
Grn	G6		Video	How to Simplify Algebraic Expressions Grade 7 Math youtu.be/BqOf_1PI_lo
Grn	G6		WS	Corbettmaths — Expanding Brackets corbettmaths.com/.../Expanding-Brackets-pdf.pdf
Grn	G6		WS	Corbettmaths — Expanding Brackets (Further) corbettmaths.com/.../expanding-brackets-pdf1.pdf

Corbettmaths worksheets (c) Corbettmaths. Free to access and print.

Red band | R1

Multiplication 1

Corbettmaths Practice Worksheet | www.corbettmaths.com

Multiplication 1

Videos 199 and 200 on www.corbettmaths.com

Examples

Workout



Click here



Scan here

Question 1: Work out the following multiplications

- | | | | | |
|-------------------|-------------------|-------------------|-------------------|-------------------|
| (a) 32×3 | (b) 15×5 | (c) 23×4 | (d) 19×3 | (e) 47×2 |
| (f) 6×21 | (g) 35×5 | (h) 59×4 | (i) 7×28 | (j) 62×6 |
| (k) 74×5 | (l) 53×9 | (m) 7×66 | (n) 83×8 | (o) 96×9 |

Question 2: Work out the following multiplications

- | | | | |
|--------------------|---------------------|---------------------|---------------------|
| (a) 223×2 | (b) 132×3 | (c) 124×4 | (d) 5×135 |
| (e) 403×6 | (f) 263×6 | (g) 365×7 | (h) 308×9 |
| (i) 6×555 | (j) 758×4 | (k) 642×8 | (l) 383×7 |
| (m) 798×9 | (n) 1294×5 | (o) 2074×6 | (p) 8×4868 |

Apply

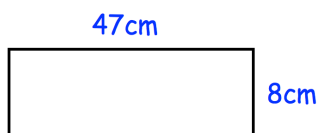
Question 1: Work out the product of 18 and 4

Question 2: How many days are there in 35 weeks?

Question 3: Mr Burns wants to buy every student in Year 11 a doughnut.
There are 120 students in Year 11.
Mr Burns buys 26 bags of doughnuts and there are 5 doughnuts in a bag.
Has Mr Burns bought enough doughnuts?

Question 4: Claudia saves £8 every month.
How much money does she save over 2 years?

Question 5: Find the area of this rectangle.



Multiplication 1

Videos 199 and 200 on www.corbettmaths.com

- Question 6: At a wedding, there are 16 tables.
15 tables seat 6 guests
1 table will seat 8 guests

Work out the total number of chairs needed.

- Question 7: Here is part of Olive's gas bill.
Each unit of gas costs 9p.

Old reading	1695 units
New reading	2104 units

Work out how much Olive will have to pay.

- Question 8: Leanne works in a cinema.
She is paid £7 per hour for the first 120 hours she works each month.
Leanne is paid an overtime rate of £9 per hour for any additional hours.
In September she works 138 hours.
Work out how much Leanne is paid.

- Question 9: Below are two boxes that contain numbers.

Box 1	Box 2												
<table><tr><td>9</td><td>2</td></tr><tr><td>3</td><td>7</td></tr><tr><td>4</td><td></td></tr></table>	9	2	3	7	4		<table><tr><td>33</td><td></td></tr><tr><td></td><td>25</td></tr><tr><td>63</td><td>94</td></tr></table>	33			25	63	94
9	2												
3	7												
4													
33													
	25												
63	94												

Choose one number from each box that multiply together to give an answer between 400 and 500.

- Question 10: Nicole owns a clothes shop.
She buys 8 jackets for £73 each.
Nicole sells the jackets for £125 each.

Work out her profit.

- Question 11: 800 people attended a charity football match between Ballymena United and AFC Telford.
Adult tickets are £9 and child tickets are £4.
Out of the 800 people at the match, 155 are children

How much money was raised for charity?

Multiplication 1

Videos 199 and 200 on www.corbettmaths.com

Question 12: Mr and Mrs Neill book a 10 day holiday in July.
They have three children.
Work out the total cost.



CORBETTMATHS HOLIDAYS

Price per day

Dates	Per adult	Per child
1 st March to 30 th April	£23	£9
1 st June to 31 st August	£26	£13

There is a £15 booking charge for every holiday

Question 13: Bertie wants to buy a table and six chairs.
Where should he buy them?

Furniture World
Table £180
Each chair £45

Furniture Land
Table Free if you buy 6 chairs
Each chair £77

Home of Furniture
Table £194
Two chairs £82

Question 14: Place the digits 4, 5, 6 and 8 into the boxes below so that

- (a) You find the largest possible answer.
(b) You find the smallest possible answer

×

Question 15: Donald is buying ribbon to wrap his Christmas presents.
The ribbon costs £1.89 per metre.
He buys 7 metres of ribbon.
Work out the total cost.



Answers



Click here



Scan here

Red band | R3

Multiples, Factors and Primes (Exam Style)

Corbettmaths Practice Worksheet | www.corbettmaths.com

Name: _____

Exam Style Questions

Multiples
Factors
Prime Numbers



Corbettmaths

Ensure you have: Pencil, pen, ruler, protractor, pair of compasses and eraser

You may use tracing paper if needed

Guidance

1. Read each question carefully before you begin answering it.
2. Don't spend too long on one question.
3. Attempt every question.
4. Check your answers seem right.
5. Always show your workings

Revision for this topic

www.corbettmaths.com/contents

Video 220

Video 216

Video 225



1. Write down the first **five** multiples of 3.

.....
(1)

2. From the list of numbers

7 9 12 21 23 30 36 45

- (a) write down the multiples of 7.

.....
(2)

- (b) write down the multiples of 5.

.....
(2)

- (c) write down the multiples of 12.

.....
(2)

3. (a) Write down two multiples of 4.

..... and
(1)

- (b) Write down two multiples of 9.

..... and
(1)

- (c) Write down a number which is a multiple of both 4 and 9.

.....
(1)

4. Write down all the factors of 16.

.....
(2)

5. Write down all the factors of 26.

.....
(2)

6. From the list of numbers

3 5 7 9 11 15 24

(a) Write down a factor of 12

.....
(1)

(b) Write down a factor of 28

.....
(1)

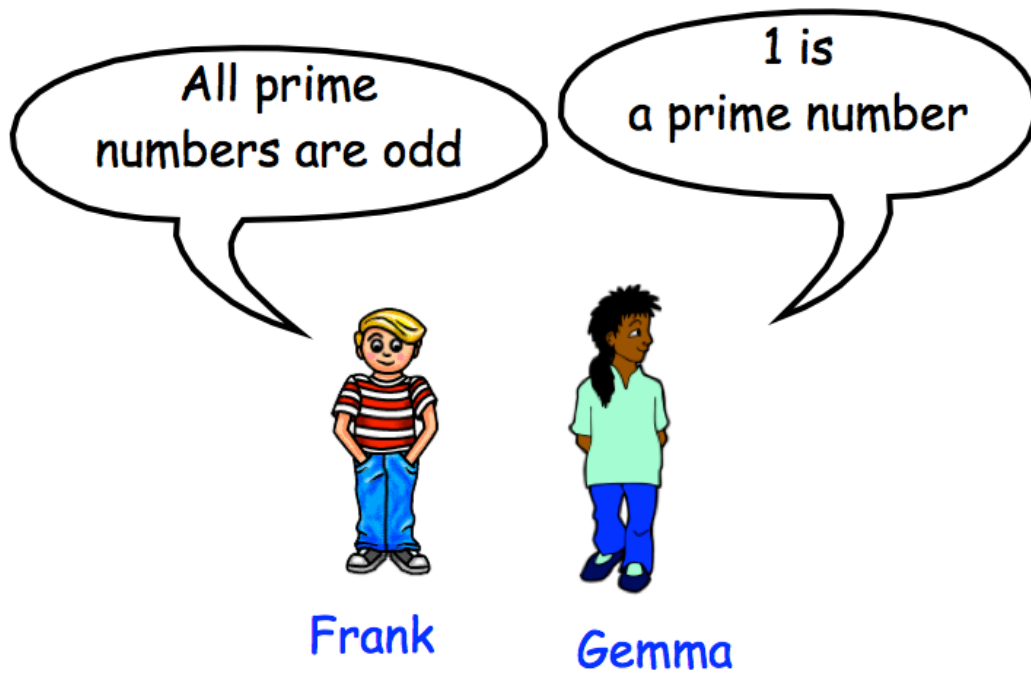
(c) Write down a factor of 81

.....
(1)

7. Write down all the prime numbers between 10 and 20.

.....
(2)

8.



Give a reason why each child is wrong.

Frank:
.....

Gemma:
.....

(2)

9. Here is a list of numbers

6 10 11 16 24 30 40

(a) Write down a multiple of 20

.....
(1)

(b) Write down a factor of 12

.....
(1)

(c) Write down a prime number

.....
(1)

10. Here is a list of 8 numbers.

15 16 17 18 20 22 24 29

(a) Write down a prime number

.....
(1)

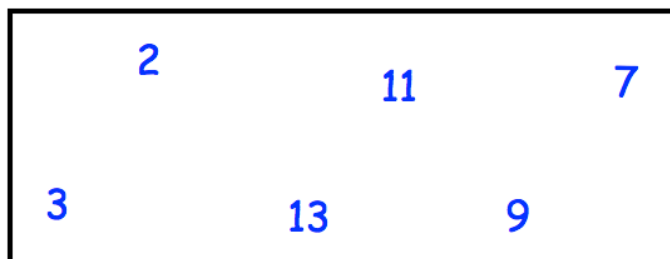
(b) Write down a factor of 30

.....
(1)

(c) Write down a multiple of 3, which is even.

.....
(1)

11.



From the numbers in the rectangle,

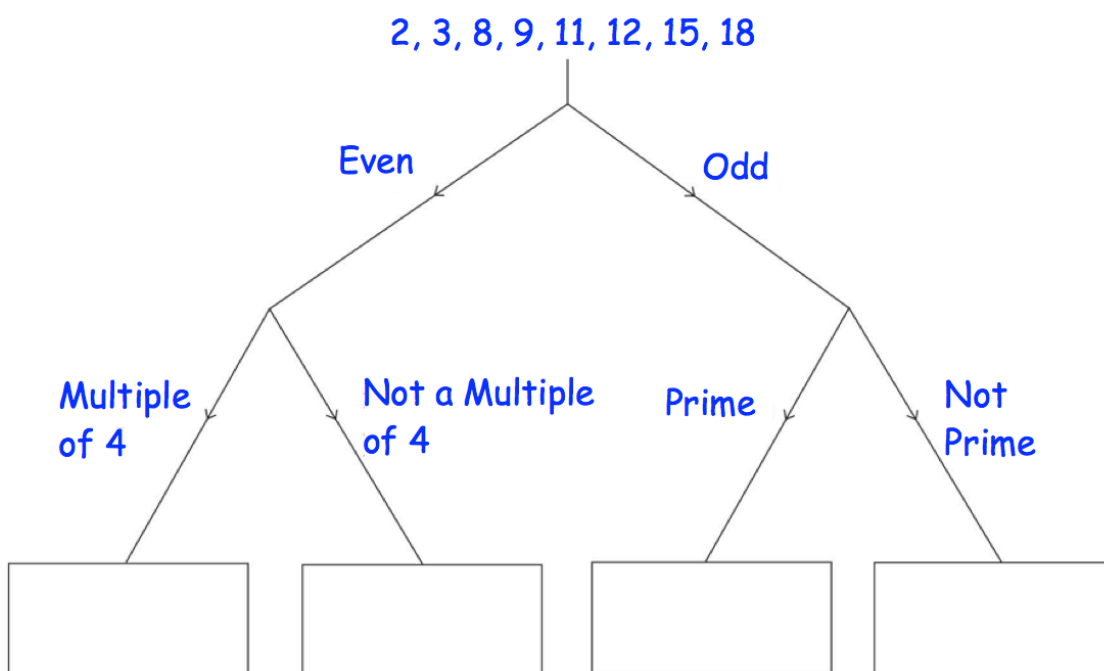
(a) Write down a factor of 35.

.....
(1)

(b) Write down the number which is **not prime**

.....
(1)

12. Sort **all** the numbers into the correct boxes.



(3)

13. A bus to Belfast leaves Antrim Bus Station every 25 minutes.
The first bus each day leaves at 7am.
Darren wants to get a bus after 8am.

What time is the first suitable bus?

.....
(2)

14. A blue light flashes every 8 seconds.
A red light flashes every 12 seconds.

Both lights have just flashed together.

After how many seconds will both lights flash together?

.....
(2)

15. Kelly is organising a barbecue.
She needs bread rolls and burgers.

Bread rolls are sold in packs of 20.
Burgers are sold in packs of 12.

Kelly buys exactly the same number of bread rolls as burgers.

What is the least number of each pack that Kelly buys?

..... packs of bread rolls

..... packs of burgers

(3)

16.

All even numbers
greater than two, can be written
as the sum of two prime numbers



Peter

Give **three** examples that show that Peter is correct.

..... + =

..... + =

..... + =

(3)

Red band | R4

Factors, Multiples and Primes

Corbettmaths Practice Worksheet | www.corbettmaths.com

Name: _____

Exam Style Questions

Multiples

Factors

Prime Numbers



Ensure you have: Pencil, pen, ruler, protractor, pair of compasses and eraser

You may use tracing paper if needed

Guidance

1. Read each question carefully before you begin answering it.
2. Don't spend too long on one question.
3. Attempt every question.
4. Check your answers seem right.
5. Always show your workings

Revision for this topic

www.corbettmaths.com/contents

Video 220

Video 216

Video 225



1. Write down the first **five** multiples of 3.

3 6 9 12 15
(1)

2. From the list of numbers

7 9 12 21 23 30 36 45

- (a) write down the multiples of 7.

7, 21
(2)

- (b) write down the multiples of 5.

30, 45
(2)

- (c) write down the multiples of 12.

12, 36
(2)

3. (a) Write down two multiples of 4.

4 and 8
(1)

- (b) Write down two multiples of 9.

9 and 18
(1)

- (c) Write down a number which is a multiple of both 4 and 9.

36
(1)

4. Write down all the factors of 16.

1, 2, 4, 8, 16
.....
(2)

5. Write down all the factors of 26.

1, 2, 13, 26
.....
(2)

6. From the list of numbers

3 5 7 9 11 15 24

- (a) Write down a factor of 12

3
.....
(1)

- (b) Write down a factor of 28

7
.....
(1)

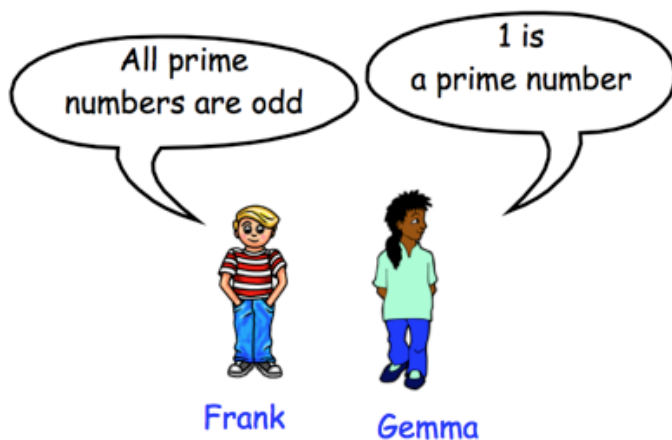
- (c) Write down a factor of 81

9
.....
(1)

7. Write down all the prime numbers between 10 and 20.

11, 13, 17, 19
(2)

- 8.



Give a reason why each child is wrong.

Frank: 2 is prime and it is even

Gemma: 1 is not prime.
A prime has two factors: 1 and itself.
(2)

9. Here is a list of numbers

6 10 11 16 24 30 40

(a) Write down a multiple of 20

40
.....
(1)

(b) Write down a factor of 12

6
.....
(1)

(c) Write down a prime number

11
.....
(1)

10. Here is a list of 8 numbers.

15 16 17 18 20 22 24 29

(a) Write down a prime number

17 (or 29)
.....
(1)

(b) Write down a factor of 30

15
.....
(1)

(c) Write down a multiple of 3, which is even.

24 (or 18)
.....
(1)

11.



From the numbers in the rectangle,

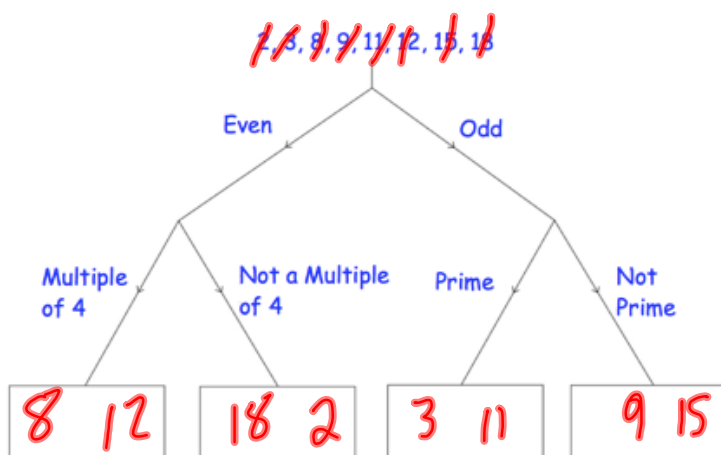
(a) Write down a factor of 35.

7
(1)

(b) Write down the number which is **not prime**

9
(1)

12. Sort **all** the numbers into the correct boxes.



(3)

13. A bus to Belfast leaves Antrim Bus Station every 25 minutes.
The first bus each day leaves at 7am.
Darren wants to get a bus after 8am.

What time is the first suitable bus?

25
50
(75) mins → 1 hour 5 minutes 8:15am
.....
(2)

14. A blue light flashes every 8 seconds.
A red light flashes every 12 seconds.

Both lights have just flashed together.

After how many seconds will both lights flash together?

8 16 24
12 24
24
.....
(2)

15. Kelly is organising a barbecue.
She needs bread rolls and burgers.

Bread rolls are sold in packs of 20.

Burgers are sold in packs of 12.

Kelly buys exactly the same number of bread rolls as burgers.

What is the least number of each pack that Kelly buys?

20 40 60
12 24 36 48 60
3
..... packs of bread rolls
5
..... packs of burgers
(3)

16.

All even numbers
greater than two, can be written
as the sum of two prime numbers



Peter

Give **three** examples that show that Peter is correct.

$$40 = 17 + 23$$

$$22 = 3 + 19$$

$$8 = 3 + 5$$

$$\begin{array}{r} 17 \quad 23 \quad 40 \\ \dots + \dots = \dots \end{array}$$

$$\begin{array}{r} 3 \quad 19 \quad 22 \\ \dots + \dots = \dots \end{array}$$

$$\begin{array}{r} 3 \quad 5 \quad 8 \\ \dots + \dots = \dots \end{array}$$

(3)

Red band | R5

Multiplication 2

Corbettmaths Practice Worksheet | www.corbettmaths.com

Multiplication 2

Videos 199 and 200 on www.corbettmaths.com

Examples

Workout



Click here



Scan here

Question 1: Work out the following multiplications

- | | | | |
|--------------------|--------------------|--------------------|--------------------|
| (a) 14×13 | (b) 23×12 | (c) 35×13 | (d) 19×15 |
| (e) 17×17 | (f) 34×23 | (g) 19×32 | (h) 48×16 |
| (i) 53×27 | (j) 44×25 | (k) 57×30 | (l) 62×16 |

Question 2: Work out the following multiplications

- | | | | |
|--------------------|--------------------|--------------------|--------------------|
| (a) 48×47 | (b) 65×78 | (c) 83×56 | (d) 74×86 |
| (e) 92×66 | (f) 73×89 | (g) 94×84 | (h) 97×69 |
| (i) 78×93 | (j) 88×88 | (k) 96×92 | (l) 76×67 |

Question 3: Work out the following multiplications

- | | | | |
|---------------------|---------------------|---------------------|---------------------|
| (a) 142×13 | (b) 127×15 | (c) 133×12 | (d) 186×11 |
| (e) 158×24 | (f) 193×25 | (g) 108×17 | (h) 170×23 |
| (i) 324×19 | (j) 405×15 | (k) 522×34 | (l) 604×28 |

Question 4: Work out the following multiplications

- | | | | |
|---------------------|---------------------|---------------------|---------------------|
| (a) 783×55 | (b) 924×46 | (c) 888×64 | (d) 606×94 |
| (e) 811×97 | (f) 777×66 | (g) 890×83 | (h) 952×74 |
| (i) 939×84 | (j) 686×58 | (k) 660×98 | (l) 968×79 |

Question 5: Work out the following multiplications

- | | | | |
|----------------------|----------------------|----------------------|----------------------|
| (a) 135×122 | (b) 142×104 | (c) 144×113 | (d) 205×147 |
| (e) 240×108 | (f) 216×156 | (g) 317×223 | (h) 562×275 |
| (i) 617×392 | (j) 384×256 | (k) 687×749 | (l) 983×991 |

Multiplication 2

Videos 199 and 200 on www.corbettmaths.com

Apply

- Question 1: There are 12 cupcakes in a box.
Dara is organising a party and wants 200 cupcakes.
He buys 16 boxes.
Does Dara have enough cupcakes?



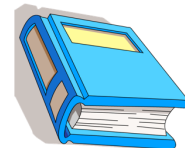
- Question 2: Find the product of 62 and 51

- Question 3: A rugby team brought 18 coaches of supporters to a cup match.
Each coach holds 53 passengers.



How many supporters are brought to the cup match by the 18 coaches?

- Question 4: Felicity spends 25 minutes reading every day.
How long does she spend reading during the month of May?



- Question 5: A cinema has 26 seats in each row.
There are 18 rows.
During a showing of movie, there are 70 empty seats.



Work out how many people watch the movie.

- Question 6: Miss Jenkins owns an electronics shop.
She order 27 laptops at £413 each.
Miss Jenkins sells the 27 laptops for £600 each.

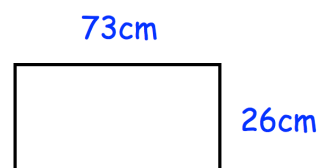
Work out the profit.

- Question 7: How many hours are there in one leap year?



- Question 8: A ruler costs 49p.
Work out the cost of 125 rulers.
Give your answer in pounds.

- Question 9: Find the area of this rectangle



Multiplication 2

Videos 199 and 200 on www.corbettmaths.com

Question 10: (a) Work out the following multiplications

$$\begin{array}{l} 1 \times 1 \\ 11 \times 11 \\ 111 \times 111 \\ 1111 \times 1111 \end{array}$$

- (b) Predict the answer to 11111×11111
 (c) Predict the answer to 11111111×11111111
 (d) When will the pattern end?

Question 11: Jenny bought a motorbike.

She paid a deposit of £345 and 36 monthly payments of £44
 At the end of the payments, she sold the motorbike for £1400.



How much did it cost Jenny in total?

- Question 12: (a) Choose any two digit number and multiply it by 13
 (b) Multiply your answer by 7
 (c) Finally multiply your new answer by 11
 (d) What do you notice?
 Can you explain why?

Question 13: Kyle is organising a charity concert at school.
 The concert is sold out.
 The hall holds 35 rows of 42 seats.
 Each person will pay £7
 How much money will Kyle raise for charity?



Question 14: Two whole numbers multiply together to give an answer of 600.
 Neither of the numbers contain the digit zero.
 What are the two numbers?

Question 15: Martina has 24 crates of oranges.
 Each crate weighs 37.3kg.
 Martina's van can hold up to 900kg.
 Will the van be able to carry all 24 crates?



Answers



Click here



Scan here

Yellow band | Y1

Division

Corbettmaths Practice Worksheet | www.corbettmaths.com

Examples



Click here



Scan here

Workout

Question 1: Work out the answers to the following divisions

- | | | | |
|-----------------|-----------------|-----------------|-----------------|
| (a) $84 \div 4$ | (b) $52 \div 2$ | (c) $72 \div 3$ | (d) $75 \div 5$ |
| (e) $54 \div 3$ | (f) $68 \div 4$ | (g) $90 \div 5$ | (h) $84 \div 6$ |
| (i) $91 \div 7$ | (j) $81 \div 3$ | (k) $87 \div 3$ | (l) $92 \div 4$ |

Question 2: Work out the answers to the following divisions

- | | | | |
|------------------|------------------|------------------|------------------|
| (a) $236 \div 2$ | (b) $156 \div 3$ | (c) $108 \div 4$ | (d) $235 \div 5$ |
| (e) $260 \div 4$ | (f) $222 \div 3$ | (g) $545 \div 5$ | (h) $312 \div 6$ |
| (i) $438 \div 6$ | (j) $171 \div 9$ | (k) $584 \div 8$ | (l) $553 \div 7$ |
| (m) $981 \div 9$ | (n) $856 \div 4$ | (o) $801 \div 9$ | (p) $406 \div 7$ |

Question 3: Work out the answers to the following divisions

- | | | | |
|-------------------|-------------------|--------------------|--------------------|
| (a) $2735 \div 5$ | (b) $3312 \div 4$ | (c) $2664 \div 3$ | (d) $6540 \div 5$ |
| (e) $3360 \div 7$ | (f) $4902 \div 6$ | (g) $7128 \div 9$ | (h) $9020 \div 5$ |
| (i) $8208 \div 8$ | (j) $7500 \div 6$ | (k) $15462 \div 3$ | (l) $24353 \div 7$ |

Question 4: Work out each of the following

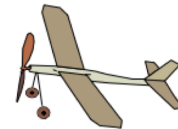
- | | | | |
|--------------------|---------------------|--------------------|--------------------|
| (a) $154 \div 11$ | (b) $192 \div 12$ | (c) $195 \div 13$ | (d) $345 \div 15$ |
| (e) $374 \div 22$ | (f) $416 \div 16$ | (g) $385 \div 11$ | (h) $648 \div 12$ |
| (i) $1150 \div 25$ | (j) $805 \div 35$ | (k) $1196 \div 52$ | (l) $630 \div 18$ |
| (m) $5580 \div 90$ | (n) $2520 \div 105$ | (o) $1755 \div 65$ | (p) $2904 \div 33$ |

Question 5: Work out each of the following.
Give each answer as a decimal.

- (a) $82 \div 4$ (b) $75 \div 6$ (c) $12 \div 5$ (d) $4 \div 5$
(e) $88 \div 5$ (f) $118 \div 8$ (g) $174 \div 12$ (h) $745 \div 20$
(i) $3406 \div 8$ (j) $4268 \div 6$ (k) $8519 \div 14$ (l) $1854 \div 24$

Apply

Question 1: A toy costs £6.
Over a week, a shop makes £162 from selling the toy.
How many toys were sold?



Question 2: A group of 3 friends take a journey in a taxi.
The total cost of the journey is £72.
The friends share the cost equally.
How much does each person pay?



Question 3: A bookshelf in a classroom is 112cm long.
A teacher has 30 mathematics textbooks, each 4cm wide.



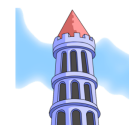
- (a) Can the teacher place all 30 textbooks on the shelf?
(b) What is the maximum number of textbooks that will fit on the shelf?

Question 4: A journey lasts 119 days.
How many weeks is this?

Question 5: Sally is paid £8 per hour.
In one week she is paid £264.
How many hours did Sally work?

Question 6: A school has 5 year groups and 835 students in total.
Each year group has an equal number of students.
How many students are in each year group?

Question 7: A group of 9 friends go on a coach tour.
The total cost for the tour is £648.
Work out the cost per person.



Question 8: The product of Jack's age and Florence's age is 266.
Jack is 14 years old.
How old is Florence?

Question 9: At a conference there are 621 people.
Each table seats 8 people.
How many tables are needed?

Question 10: Daisy is buying rulers.
She has £10.
Each ruler costs 74p
Daisy buys as many rulers as she can.

- (a) How many rulers does Daisy buy?
- (b) How much change should Daisy receive?

Question 11: Miss Jenkins has 18 bags of sweets.
Each bag contains 30 sweets.
Miss Jenkins shares as many sweets as possible equally among the 16 students in her class.



- (a) How many sweets does each student receive?
- (b) How many sweets are left over?

Question 12: Harry hires a car from Holiday Cars for 3 days.
His total bill was £204.
How many miles did Harry drive?

Holiday Cars

£45 per day
20p per mile

Question 13: Leah bought a new car costing £18,000
She paid a deposit of £2,000.
Leah paid the rest of the money over 50 equal monthly payments.
How much was each monthly payment?

Question 14: James hired a holiday cottage for 7 days for £406
Ben hires the same cottage, at the same price per day, for 10 days.
How much will this cost Ben?

Question 15: The product of three numbers is 1001.
The first two numbers are 7 and 11.
What is the third number?

Question 16: There are 1560 sweets in a tub.
Katherine and her friends share the sweets equally.
Each person receives 65 sweets.
How many friends does Katherine have?

Question 17: A theatre has 28 seats in each row.
There are 1036 seats in total.
How many rows are there?

Answers



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Yellow band | Y2

Division: Remainders

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Examples

Workout



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Question 1: Work out the remainder for each of the following divisions.

- | | | | | |
|------------------|------------------|------------------|-------------------|------------------|
| (a) $17 \div 2$ | (b) $23 \div 5$ | (c) $14 \div 3$ | (d) $19 \div 4$ | (e) $14 \div 2$ |
| (f) $26 \div 6$ | (g) $45 \div 10$ | (h) $31 \div 7$ | (i) $12 \div 9$ | (j) $30 \div 8$ |
| (k) $40 \div 11$ | (l) $52 \div 8$ | (m) $49 \div 5$ | (n) $66 \div 9$ | (o) $80 \div 7$ |
| (p) $102 \div 5$ | (q) $79 \div 3$ | (r) $139 \div 4$ | (s) $283 \div 10$ | (t) $90 \div 12$ |

Question 2: Work out the remainder for each of the following divisions.

- | | | | |
|-------------------|-------------------|-------------------|-------------------|
| (a) $326 \div 7$ | (b) $776 \div 3$ | (c) $359 \div 6$ | (d) $232 \div 8$ |
| (e) $400 \div 12$ | (f) $452 \div 15$ | (g) $377 \div 16$ | (h) $283 \div 11$ |

Question 3: Work out the following divisions.

Give each answer as a decimal number. e.g. $13 \div 2 = 6.5$

- | | | | | |
|-------------------|--------------------|--------------------|-------------------|------------------|
| (a) $7 \div 2$ | (b) $9 \div 5$ | (c) $43 \div 2$ | (d) $27 \div 5$ | (e) $86 \div 5$ |
| (f) $56 \div 10$ | (g) $14 \div 4$ | (h) $66 \div 4$ | (i) $51 \div 6$ | (j) $41 \div 4$ |
| (k) $75 \div 4$ | (l) $26 \div 8$ | (m) $38 \div 8$ | (n) $40 \div 3$ | (o) $29 \div 3$ |
| (p) $123 \div 15$ | (q) $111 \div 12$ | (r) $300 \div 9$ | (s) $748 \div 20$ | (t) $253 \div 6$ |
| (u) $853 \div 40$ | (v) $1879 \div 20$ | (w) $8161 \div 80$ | | |

Question 4: Work out the following divisions.

Give each answer as a mixed number and simplify any fraction. e.g. $14 \div 3 = 4\frac{2}{3}$

- | | | | | |
|-------------------|-------------------|-------------------|-------------------|-----------------|
| (a) $11 \div 2$ | (b) $38 \div 3$ | (c) $11 \div 6$ | (d) $43 \div 10$ | (e) $21 \div 5$ |
| (f) $54 \div 10$ | (g) $50 \div 8$ | (h) $45 \div 7$ | (i) $78 \div 5$ | (j) $99 \div 4$ |
| (k) $155 \div 6$ | (l) $290 \div 11$ | (m) $481 \div 12$ | (n) $324 \div 5$ | (o) $83 \div 9$ |
| (p) $384 \div 15$ | (q) $772 \div 10$ | (r) $358 \div 20$ | (s) $475 \div 40$ | |

Apply

Question 1: Simon is sharing 27 marbles equally between 4 friends.
How many marbles are left over?

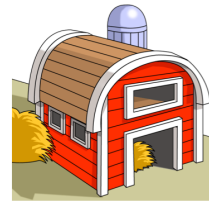


Question 2: Rebecca is selling raffle tickets in booklets of 5.
She has 86 raffle tickets.

- (a) How many booklets can Rebecca sell?
- (b) How many tickets will be left over?

Question 3: Eggs are being packed into boxes of 6.
Farmer Richards has 77 eggs.

- (a) How many boxes can he fill?
- (b) How many eggs will be left over?



Question 4: At a wedding there are 125 guests.
8 people can sit at each table.
All the tables are filled, except one.
How many guests sit at the table that is not filled?

Question 5: Burt is making cupcakes.
He places the cupcakes in boxes of 12.
Burt has 200 cupcakes.
How many boxes can he fill?



Question 6: Five friends share £13.
How much do they receive each?

Question 7: The perimeter of a square is 171cm
Find the length of each side.

Question 8: Michael is saving for a holiday.
The holiday will cost him £885.
He will save money for one year.
Michael will save an equal amount each month to pay for the holiday.
How much should he save each month?



Answers



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Yellow band | Y3

Order of Operations (BODMAS)

Corbettmaths Practice Worksheet | www.corbettmaths.com

Name: _____

Exam Style Questions



Order of Operations

Corbettmaths

Ensure you have: Pencil, pen, ruler, protractor, pair of compasses and eraser

You may use tracing paper if needed

Guidance

1. Read each question carefully before you begin answering it.
2. Don't spend too long on one question.
3. Attempt every question.
4. Check your answers seem right.
5. Always show your workings

Revision for this topic

Secondary

Video 211



1. Calculate

(a) $16 - 5 \times 2$

.....
(1)

(b) $10 - 3^2$

.....
(1)

(c) $5 \times (2 + 3)$

.....
(1)

2. Calculate

(a) $10 + 3 \times 2$

.....
(1)

(b) $8 \div 2 + 12 \div 4$

.....
(1)

(c) $3 \times 10 \div 5 - 1$

.....
(1)

3. Calculate

(a) $6 + 6 \div 3$

.....
(1)

(b) $8 + 3(5 - 1)$

.....
(1)

(c) $9 \times 2 + 20 \div 2$

.....
(1)

4. Put brackets in the following statements to make them true

(a) $6 \times 7 + 3 - 8 = 52$

(1)

(b) $4 + 3 \times 7 - 1 = 42$

(1)

5. Work out

(a) $14 + 12 \div 2$

.....
(1)

(b) $6 \times 4 - 7 \times 3$

.....
(1)

6. Work out

(a) $2^3 + 3^2$

.....
(1)

(b) $2^2 \times 3^3$

.....
(2)

7. Work out $(2 + 5)^2$

.....
(1)

8. Work out

(a) $(2 + 5)^2$

.....
(1)

(b) $5 + 3 \times 6$

.....
(1)

(c) $22 - 14 \div 2$

.....
(1)

(d) $(9 + 4) \times (100 \div 25)$

.....
(1)

(e) $7 \times 5 - 10$

.....
(1)

9. Joey thinks the answer to $16 + 4 \times 2$ is 40.
Albert thinks the answer to $16 + 4 \times 2$ is 24.

Who is correct?
Explain your answer.

(2)

10. Work out

(a) $4 \times (3 + 17)$

.....
(1)

(b) $10 - 2 \times 5$

.....
(1)

(c) $30 - 5 \times 2$

.....
(1)

Yellow band | Y3

Order of Operations Exercise

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Order of Operations (BODMAS)

Video 211 on Corbettmaths

Examples

Workout



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Scan here

Question 1: Work out

- | | | | |
|---------------------------|----------------------------|-------------------------|--------------------------|
| (a) $7 + 2 \times 3$ | (b) $9 + 4 \times 2$ | (c) $10 + 2 \times 2$ | (d) $18 + 4 \div 2$ |
| (e) $20 - 5 \times 2$ | (f) $8 - 2 \times 3$ | (g) $21 - 9 \div 3$ | (h) $100 - 40 \times 2$ |
| (i) $16 \div 1 - 3$ | (j) $5 + 5 \times 5$ | (k) $13 - 7 \div 1$ | (l) $7 \times 6 - 4$ |
| (m) $9 + 3 - 2$ | (n) $20 - 5 + 6$ | (o) $21 - 17 + 4$ | (p) $30 \times 4 \div 2$ |
| (q) $(7 + 7) \div 2$ | (r) $35 - (9 + 3)$ | (s) $40 \times (2 + 3)$ | (t) $60 \div (1 + 5)$ |
| (u) $15 \div (3 + 2)$ | (v) $9 \times (7 + 4)$ | (w) $90 \div (52 - 7)$ | (x) $(8 + 9) \times 3$ |
| (y) $10 + 5 + 3 \times 3$ | (z) $100 - 6 + 2 \times 3$ | | |

Question 2: Work out

- | | | | |
|----------------------|-----------------------|----------------------|-----------------------------|
| (a) $5 - 2^2$ | (b) $7 + 3^2$ | (c) $9^2 + 1$ | (d) $6^2 - 5^2$ |
| (e) $(7 - 2)^2$ | (f) $(4 + 3)^2$ | (g) $(1 + 2)^3$ | (h) $(2 + 8)^3$ |
| (i) $10 - \sqrt{16}$ | (j) $\sqrt{(2 + 14)}$ | (k) $\sqrt{4 + 3^2}$ | (l) $2 \times 5 - \sqrt{4}$ |

Question 3: Work out

- | | | | |
|-------------------------------|------------------------------|------------------------------|----------------------------|
| (a) $5 \times 3 + 2 \times 6$ | (b) $9 \div 3 + 15 \times 2$ | (c) $10 \div 2 - 2 \times 1$ | (d) $5 \times (2 + 1) + 4$ |
| (e) $8 + (5 - 1) \times 3$ | (f) $50 - (1 + 4) \times 4$ | (g) $19 \times 2 + 5^2$ | (h) $8^2 + 2 \times 3^2$ |
| (i) $7 \times (8 \div 4)^2$ | (j) $11 + 11 - 6^2 \div 2$ | | |

Question 4: Copy out the following and insert brackets in each to make the correct answer.

- | | | |
|-------------------------------|------------------------------------|-------------------------------------|
| (a) $10 \times 2 + 6 = 80$ | (b) $5 + 5 \div 5 = 2$ | (c) $18 - 6 \div 2 = 6$ |
| (d) $5 + 2 \times 3 + 1 = 13$ | (e) $2 \times 7 + 1 \times 3 = 48$ | (f) $9 + 3^2 \times 10 \div 2 = 90$ |

Order of Operations (BODMAS)

Video 211 on Corbettmaths

Apply

Question 1: Matthew says $9 + 3 \times 2 = 15$. Is he correct?

Question 2: Samuel says $6 + 4 \times 9 = 90$. Is he correct?

Question 3: Using the numbers 2, 3 and 4 and the operations +, −, and $\times$ make as many different possible answers.

Question 4: Matilda thinks of a number, n .
She adds 2 and then multiplies by 3.

Which expression below is correct?

A

$$n + 2 \times 3$$

B

$$3n + 2$$

C

$$(n + 2) \times 3$$

Question 5: Can you spot any mistakes?

Work out $9 + 4 \times 3 + 2$

$$= 13 \times 3 + 2$$

$$= 39 + 2$$

$$= 41$$

Extension Task

Using four number 2's try to make as many different answers as you can.
You may use +, −, $\times$, $\div$ and brackets.

You may use one or more of the 2's as powers.

Answers



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Yellow band | Y4

Division Exam Style Questions

Corbettmaths Practice Worksheet | www.corbettmaths.com

Name: _____

Exam Style Questions



Negatives: addition, subtraction,
multiplication and division

Corbettmaths

Ensure you have: Pencil, pen, ruler, protractor, pair of compasses and eraser

You may use tracing paper if needed

Guidance

1. Read each question carefully before you begin answering it.
2. Don't spend too long on one question.
3. Attempt every question.
4. Check your answers seem right.
5. Always show your workings

Revision for this topic

www.corbettmaths.com/contents

Video 205

Video 206

Video 207



1. Write down the number that is

(a) 4 less than 1

.....
(1)

(b) 2 more than -8

.....
(1)

(c) -2 multiplied by 7

.....
(1)

(d) -8 multiplied by -3

.....
(1)

(e) -10 divided by 5

.....
(1)

(f) -12 divided by -2

.....
(1)

2. Write down the answer to

(a) $3 - 7$

.....
(1)

(b) -2×-7

.....
(1)

(c) $21 \div -7$

.....
(1)

(d) $-100 \div -4$

.....
(1)

3. Calculate

(a) $(-54) \div (-9)$

.....
(1)

(b) $10 - (-2)$

.....
(1)

4. Calculate

(a) 3×-13

.....
(1)

(b) $-48 \div -8$

.....
(1)

5. Complete the following.

(a)

$$20 \div \boxed{} = -4$$

(1)

(b)

$$-7 - \boxed{} = -10$$

(1)

(c)

$$5 + \boxed{} = 0$$

(1)

(d)

$$-6 \times \boxed{} = 36$$

(1)

6. Work out each of the following

(a) $5 - 8 + 1$

.....
(1)

(b) $7 - 9 - 2$

.....
(1)

(c) $-10 + 3 + 13$

.....
(1)

7. Work out each of the following

(a) $7 - (-4)$

.....
(1)

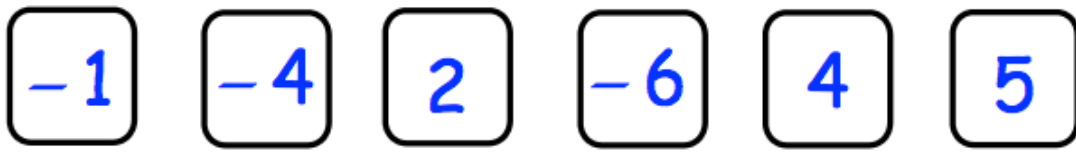
(b) $-2 + (-3)$

.....
(1)

(c) $-6 - (-8)$

.....
(1)

8. Emily has the following cards



Emily is going to choose two cards and multiply the numbers on them.
What cards should Emily choose to make the largest possible answer?

$$\square \times \square$$

(1)

-
9. Fill in the missing numbers

(a)

$$\square + 2 = -1$$

(1)

(b)

$$8 + \square = 0$$

(1)

(c)

$$-8 - \square = 3$$

(1)

10. Fill in the missing numbers

(a)

$$\boxed{2} \times \boxed{} = \boxed{-16}$$

(1)

(b)

$$\boxed{-1} \times \boxed{} = \boxed{2}$$

(1)

(c)

$$\boxed{} \times \boxed{-7} = \boxed{42}$$

(1)

(d)

$$\boxed{} \div \boxed{3} = \boxed{-15}$$

(1)

11. Work out

(a)

$$(-3)^2$$

.....
(1)

(b)

$$(-2)^3$$

.....
(1)

Green band | G1

Negative Numbers: Mult and Division

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Examples



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Workout

Question 1: Answer each of the following multiplications

- | | | | |
|---------------------|---------------------|----------------------|----------------------|
| (a) 2×-3 | (b) -4×3 | (c) -5×5 | (d) -7×-2 |
| (e) -6×-3 | (f) 8×-4 | (g) -9×3 | (h) -5×-8 |
| (i) -9×7 | (j) 10×-8 | (k) 7×-4 | (l) 6×8 |
| (m) -11×3 | (n) 4×-15 | (o) -12×-12 | (p) -5×7 |
| (q) 9×-8 | (r) -7×-8 | (s) 12×-6 | (t) 4×-13 |
| (u) -11×10 | (v) -20×-6 | (w) 14×7 | (x) -18×-13 |
| (y) 25×-7 | (z) -16×21 | | |

Question 2: Answer each of the following multiplications

- | | | | |
|-----------------------------|------------------------------|------------------------------|------------------------------|
| (a) $2 \times 3 \times -2$ | (b) $-3 \times 2 \times 5$ | (c) $-5 \times -6 \times 2$ | (d) $10 \times -3 \times -4$ |
| (e) $-9 \times 2 \times -2$ | (f) $-4 \times -3 \times -5$ | (g) $-8 \times -8 \times -2$ | (h) $5 \times -4 \times -7$ |

Question 3: Work out each of the following

- | | | | |
|---------------|--------------|---------------|---------------|
| (a) $(-3)^2$ | (b) $(-6)^2$ | (c) $(-2)^2$ | (d) $(-1)^2$ |
| (e) $(-10)^2$ | (f) $(-8)^2$ | (g) $(-12)^2$ | (h) $(-20)^2$ |

Question 4: Work out each of the following

- | | | | |
|--------------|---------------|--------------|--------------|
| (a) $(-2)^3$ | (b) $(-3)^3$ | (c) $(-1)^3$ | (d) $(-5)^3$ |
| (e) $(-1)^4$ | (f) $(-10)^4$ | (g) $(-2)^4$ | (h) $(-3)^4$ |

Question 5: Answer each of the following divisions

- | | | | |
|------------------|------------------|------------------|------------------|
| (a) $-10 \div 2$ | (b) $-12 \div 3$ | (c) $-24 \div 4$ | (d) $-42 \div 6$ |
|------------------|------------------|------------------|------------------|

Negative Numbers: Multiplication and Division

Videos 206 and 207 on Corbettmaths

- | | | | |
|-------------------|--------------------|--------------------|--------------------|
| (e) $9 \div -3$ | (f) $21 \div -7$ | (g) $-44 \div 11$ | (h) $-72 \div 9$ |
| (i) $-10 \div -5$ | (j) $-28 \div -4$ | (k) $-30 \div -3$ | (l) $-48 \div -8$ |
| (m) $-6 \div 6$ | (n) $24 \div -3$ | (o) $-12 \div -12$ | (p) $-132 \div 11$ |
| (q) $72 \div -8$ | (r) $-108 \div -9$ | (s) $36 \div -9$ | (t) $100 \div -4$ |
| (u) $-95 \div 5$ | (v) $-49 \div -7$ | (w) $144 \div 12$ | (x) $-215 \div -5$ |
| (y) $90 \div -15$ | (z) $-342 \div 9$ | | |

Question 6: Answer each of the following divisions

- | | | | |
|----------------------|----------------------|---------------------|----------------------|
| (a) -9×-5 | (b) $-32 \div 8$ | (c) $66 \div -6$ | (d) 2×-12 |
| (e) $-24 \div -3$ | (f) -12×7 | (g) $-54 \div 6$ | (h) -16×-2 |
| (i) 8×-6 | (j) -7×-6 | (k) $40 \div -8$ | (l) $56 \div -7$ |
| (m) $-81 \div -9$ | (n) -14×-5 | (o) 10×-11 | (p) $-65 \div 5$ |
| (q) -90×-3 | (r) $-170 \div -10$ | (s) $1 \div -1$ | (t) -1.5×-3 |
| (u) $-17 \div 2$ | (v) 2.2×-10 | (w) $-93 \div -10$ | (x) -6.2×-3 |
| (y) -9×10.5 | (z) $52 \div -5$ | | |

Apply

Question 1: Work out the missing numbers

(a) $-6 \times \square = -30$

(b) $-6 \times \square = 0$

(c) $-6 \times \square = 18$

(d) $\square \times -6 = -54$

Question 2: Work out the missing numbers

(a) $-24 \div \square = 6$

(b) $\square \div -8 = -2$

(c) $32 \div \square = -4$

(d) $\square \div -3 = 4$

Negative Numbers: Multiplication and Division

Videos 206 and 207 on Corbettmaths

Question 3: Write down eight multiplications with an answer of -20

Question 4: Write down eight divisions with an answer of -3

Question 5: Write down the next two numbers in each of these number sequences

(a) 2, -6 , 18, ..., ...

(b) -5 , 10, -20 , ..., ...

(c) 240, -120 , 60, ..., ...

(d) -12 , 6, -3 , ..., ...

Question 6: Shown below is a “magic square” where the product of each row, column and diagonal are equal.

Find the missing numbers

	36	
9	6	4
-12		

Question 7: Shown below is a “magic square” where the product of each row, column and diagonal are equal.

Find the missing numbers

-5	100	
4		25
		-20

Answers



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Green band | G2

Order of Operations (BODMAS)

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Name: _____

Exam Style Questions



Order of Operations

Corbettmαths

Ensure you have: Pencil, pen, ruler, protractor, pair of compasses and eraser

You may use tracing paper if needed

Guidance

1. Read each question carefully before you begin answering it.
2. Don't spend too long on one question.
3. Attempt every question.
4. Check your answers seem right.
5. Always show your workings

Revision for this topic

Secondary

Video 211



1. Calculate

(a) $16 - 5 \times 2$

.....
(1)

(b) $10 - 3^2$

.....
(1)

(c) $5 \times (2 + 3)$

.....
(1)

2. Calculate

(a) $10 + 3 \times 2$

.....
(1)

(b) $8 \div 2 + 12 \div 4$

.....
(1)

(c) $3 \times 10 \div 5 - 1$

.....
(1)

3. Calculate

(a) $6 + 6 \div 3$

.....
(1)

(b) $8 + 3(5 - 1)$

.....
(1)

(c) $9 \times 2 + 20 \div 2$

.....
(1)

4. Put brackets in the following statements to make them true

(a) $6 \times 7 + 3 - 8 = 52$

(1)

(b) $4 + 3 \times 7 - 1 = 42$

(1)

5. Work out

(a) $14 + 12 \div 2$

.....
(1)

(b) $6 \times 4 - 7 \times 3$

.....
(1)

6. Work out

(a) $2^3 + 3^2$

.....
(1)

(b) $2^2 \times 3^3$

.....
(2)

7. Work out $(2 + 5)^2$

.....
(1)

8. Work out

(a) $(2 + 5)^2$

.....
(1)

(b) $5 + 3 \times 6$

.....
(1)

(c) $22 - 14 \div 2$

.....
(1)

(d) $(9 + 4) \times (100 \div 25)$

.....
(1)

(e) $7 \times 5 - 10$

.....
(1)

9. Joey thinks the answer to $16 + 4 \times 2$ is 40.
Albert thinks the answer to $16 + 4 \times 2$ is 24.

Who is correct?
Explain your answer.

(2)

10. Work out

(a) $4 \times (3 + 17)$

.....
(1)

(b) $10 - 2 \times 5$

.....
(1)

(c) $30 - 5 \times 2$

.....
(1)

Green band | G2

Arithmetic with Negatives

Corbettmaths Practice Worksheet | www.corbettmaths.com

Name: _____

Exam Style Questions



Negatives: addition, subtraction,
multiplication and division

Corbettmaths

Ensure you have: Pencil, pen, ruler, protractor, pair of compasses and eraser

You may use tracing paper if needed

Guidance

1. Read each question carefully before you begin answering it.
2. Don't spend too long on one question.
3. Attempt every question.
4. Check your answers seem right.
5. Always show your workings

Revision for this topic

www.corbettmaths.com/contents

Video 205

Video 206

Video 207



1. Write down the number that is

(a) 4 less than 1

.....
(1)

(b) 2 more than -8

.....
(1)

(c) -2 multiplied by 7

.....
(1)

(d) -8 multiplied by -3

.....
(1)

(e) -10 divided by 5

.....
(1)

(f) -12 divided by -2

.....
(1)

2. Write down the answer to

(a) $3 - 7$

.....
(1)

(b) -2×-7

.....
(1)

(c) $21 \div -7$

.....
(1)

(d) $-100 \div -4$

.....
(1)

3. Calculate

(a) $(-54) \div (-9)$

.....
(1)

(b) $10 - (-2)$

.....
(1)

4. Calculate

(a) 3×-13

.....
(1)

(b) $-48 \div -8$

.....
(1)

5. Complete the following.

(a)

$$20 \div \boxed{} = -4$$

(1)

(b)

$$-7 - \boxed{} = -10$$

(1)

(c)

$$5 + \boxed{} = 0$$

(1)

(d)

$$-6 \times \boxed{} = 36$$

(1)

6. Work out each of the following

(a) $5 - 8 + 1$

.....
(1)

(b) $7 - 9 - 2$

.....
(1)

(c) $-10 + 3 + 13$

.....
(1)

7. Work out each of the following

(a) $7 - (-4)$

.....
(1)

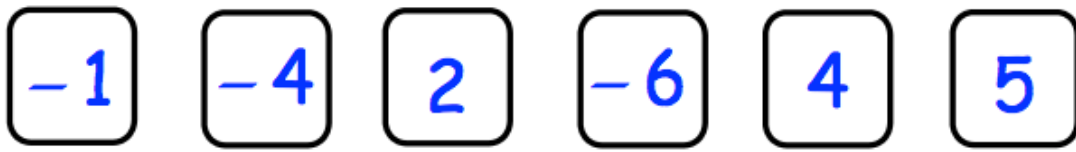
(b) $-2 + (-3)$

.....
(1)

(c) $-6 - (-8)$

.....
(1)

8. Emily has the following cards



Emily is going to choose two cards and multiply the numbers on them.
What cards should Emily choose to make the largest possible answer?

$$\square \times \square$$

(1)

-
9. Fill in the missing numbers

(a)

$$\square + 2 = -1$$

(1)

(b)

$$8 + \square = 0$$

(1)

(c)

$$-8 - \square = 3$$

(1)

10. Fill in the missing numbers

(a)

$$\boxed{2} \times \boxed{} = \boxed{-16}$$

(1)

(b)

$$\boxed{-1} \times \boxed{} = \boxed{2}$$

(1)

(c)

$$\boxed{} \times \boxed{-7} = \boxed{42}$$

(1)

(d)

$$\boxed{} \div \boxed{3} = \boxed{-15}$$

(1)

11. Work out

(a)

$$(-3)^2$$

.....
(1)

(b)

$$(-2)^3$$

.....
(1)

Green band | G3/G4

Substitution

Corbettmaths Practice Worksheet | www.corbettmaths.com

Substitution

Video 20 on www.corbettmaths.com

Examples



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Workout

Question 1: If $a = 7$ $b = 10$ $c = 3$ $d = 8$ and $e = 15$
Find the value of each expression.

- | | | | |
|-------------------|-------------------|-------------------|-------------------|
| (a) $a + 5$ | (b) $b - 4$ | (c) $c + d$ | (d) $e - d$ |
| (e) $2a$ | (f) $4b$ | (g) $3e$ | (h) $5c$ |
| (i) $\frac{b}{2}$ | (j) $\frac{e}{5}$ | (k) $\frac{d}{4}$ | (l) $\frac{a}{2}$ |
| (m) a^2 | (n) b^2 | (o) c^2 | (p) d^2 |
| (q) $2a + 1$ | (r) $3b - 7$ | (s) $9c + 11$ | (t) $4e - 45$ |
| (u) $2a + 3c$ | (v) $4d - b$ | (w) $5a + 2d$ | (x) $e - 4c$ |
| (y) $30 - 4a$ | (z) $15 - 3c$ | | |

Question 2: If $f = 5$ $g = 6$ $h = 4$ and $i = 2$
Find the value of each expression.

- | | | | |
|----------------|---------------|---------------|---------------|
| (a) fg | (b) hi | (c) fgh | (d) i^3 |
| (e) $\sqrt{h}$ | (f) $3f + 2g$ | (g) $5h + 7i$ | (h) $9h - 7i$ |

Question 3: If $a = -2$ $b = 5$ $c = -6$ $d = 10$ and $e = 9$
Find the value of each expression.

- | | | | |
|-------------------|-------------------|----------------|---------------|
| (a) $a + 4$ | (b) $b - 8$ | (c) $c + e$ | (d) $a - d$ |
| (e) $d - c$ | (f) $2c$ | (g) $7a$ | (h) $-7b$ |
| (i) $2d + 3c$ | (j) $6e + 3a$ | (k) $5a + 7$ | (l) $20 + 4a$ |
| (m) ac | (n) $40 - d$ | (o) $2e - a$ | (p) $bd + a$ |
| (q) $\frac{a}{2}$ | (r) $\frac{d}{4}$ | (s) $\sqrt{e}$ | (t) c^2 |

Substitution

Video 20 on www.corbettmaths.com

Question 4: If $a = 1.5$ $b = 4$ $c = 6$ $d = 0.5$ and $e = -3$
Find the value of each expression.

- (a) $4(a + d)$ (b) $5(c + b)$ (c) $3(10 - e)$ (d) abc
 (e) e^3 (f) d^2 (g) $5b^2$ (h) $8e^2 + 3$
 (i) $\frac{b+2}{3}$ (j) $\frac{2c-e}{4}$ (k) $\frac{10d+4b}{7}$

Question 5: $P = 2L + 2W$, work out P if $L = 8$ and $W = 3$.

Question 6: $C = 15h + 30$, work out C if $h = 6$.

Apply

Question 1: The cost of hiring a car for a number of days is calculated using the formula

$$\text{Hire Cost} = 30 \times \text{Number of Days} + 50$$



- (a) Calculate the cost of hiring a car for 4 days.
 (b) Calculate the cost of hiring a car for 9 days.
 (c) The hire cost is £110, how many days was the car hired for?
 (d) The hire cost is £380, how many days was the car hired for?

Question 2: The cost of photocopying is given as:

$$\text{Cost in pence} = 3 \times \text{number of black \& white pages} + 15 \times \text{number of colour pages}$$

- (a) Ella orders 20 black & white pages and 6 colour pages, work out the cost.
 (b) Tom orders 400 black & white pages and 70 colour pages, work out the cost.

Question 3: The time in minutes, taken to cook a chicken is given by the formula

$$\text{Time} = 40 \text{ minutes per kilogram plus } 20 \text{ minutes}$$

- (a) Work out the time taken to cook a 5kg chicken.
 (b) Work out the time taken to cook a 2.5kg chicken.



Substitution

Video 20 on www.corbettmaths.com

Question 4: This formula is used to calculate the weekly pay of a letting agent.

Weekly pay = basic pay + number of houses rented x bonus

The basic pay is £400 and a bonus of £75 is paid for each house rented.
Mrs Lewis rents out 5 houses in one week.
Calculate her pay.



Question 5: This formula can be used to convert between Celsius and Fahrenheit:

$$F = 1.8C + 32$$

- (a) Work out the value of F when C = 10
- (b) Work out the value of F when C = 20
- (c) Work out the value of F when C = 4
- (d) Work out the value of C when F = 35.6
- (e) Work out the value of C when F = 41
- (f) Work out the value of C when F = 112
- (g) Find a temperature when F and C are the same value.



Answers



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Green band | G5

Collecting Like Terms

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Examples



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Workout

Question 1: Simplify each of the following

- | | | | |
|--------------------------|--------------------------|-----------------------------------|---------------------|
| (a) $y + y + y + y$ | (b) $w + w + w + w + w$ | (c) $a + a + a + a + a + a$ | (d) $s + s + s$ |
| (e) $n + n$ | (f) $g + g + g + g - g$ | (g) $y + y + y + y - y - y$ | (h) $p + p - p - p$ |
| (i) $3y + 2y$ | (j) $4a + 3a$ | (k) $9k + 5k$ | (l) $7m + m$ |
| (m) $15c + 20c$ | (n) $6w - 3w$ | (o) $10y + 3y - 5y$ | (p) $20t - 14t$ |
| (q) $7x - 3x - x$ | (r) $8k - 8k$ | (s) $7y - 2y + y$ | (t) $5u - 4u$ |
| (u) $y^2 + y^2$ | (v) $a^2 + a^2 + a^2$ | (w) $c^2 + c^2 + c^2 + c^2 + c^2$ | (x) $7y^2 + 3y^2$ |
| (y) $2w^2 + 4w^2 + 8w^2$ | (z) $6y^2 - 2y^2 + 3y^2$ | | |

Question 2: Simplify the following expressions

- | | | | |
|--------------------------|--------------------|--------------------|-------------------|
| (a) $4u - 6u$ | (b) $8w - 9w$ | (c) $4a + 2a - 9a$ | (d) $2y - 9y$ |
| (e) $-3g - 2g$ | (f) $-4f + 9f$ | (g) $-m - 7m$ | (h) $5y^2 - 7y^2$ |
| (i) $6a^2 + 2a^2 - 9a^2$ | (j) $ab + ab + ab$ | | |

Question 3: Simplify the following expressions

- | | | |
|-----------------------------|---------------------------|--------------------------------|
| (a) $3a + 2b + 4a + b$ | (b) $7y + 5y + 2h + 2h$ | (c) $g + 8a + 2a + g$ |
| (d) $7m + 7p + 8m + p + 2p$ | (e) $9e + 2 + e + 2$ | (f) $4 + 3a + 2a + 8$ |
| (g) $2y + 4 + 3y - 1$ | (h) $8 + 3w - w - 3$ | (i) $5 - 4s - 2 + 10s$ |
| (j) $3x + 6y + 5x - 2y$ | (k) $6m - 2s + 11s + m$ | (l) $2a + 3b - 2 + a + 3b + 4$ |
| (m) $3a - 2b + a - 5b$ | (n) $2x - 2y - 6x + 5y$ | (o) $y - 4m - 3y - 5m$ |
| (p) $7p - 2q - q + 3r + 4r$ | (q) $11c + 8d - 6c - 11d$ | |

Collecting Like Terms

Video 9 on www.corbettmaths.com

Question 4: Simplify the following

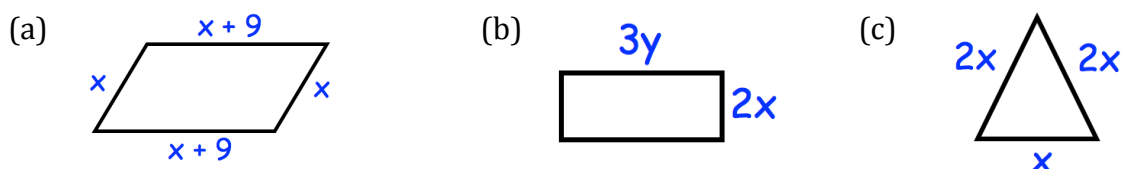
(a) $3y^2 + 4ab + 7y^2 + ab$ (b) $9x^2 - 2x - 11x^2 + 5x$ (c) $7ac - 3ab + 9ab - 7ac$

Question 5: Expand and simplify the following

(a) $2(y + 3) + 3(y + 1)$	(b) $8(x + 2) + 3(x + 3)$	(c) $4(x - 1) + 2(x + 3)$
(d) $5x + 3 + 2(x + 9)$	(e) $3(2y + 1) + 4(2y + 5)$	(f) $5(2x + 3) + 2(3x + 1)$
(g) $7(c + 2) + 3(c - 2)$	(h) $5(2a + 7) + 2(9a - 4)$	(i) $9(t + 3) + 3(2t - 11)$
(j) $2(x - 4) + 5(x - 2)$	(k) $6(y - 1) - 2(y + 3)$	(l) $8(x + 2) - 3(x - 2)$
(m) $5(2y - 3) + 3(y - 2)$	(n) $2(4w - 5) - 2(w - 7)$	(o) $5(3y + 7) - 3(2y - 5)$

Apply

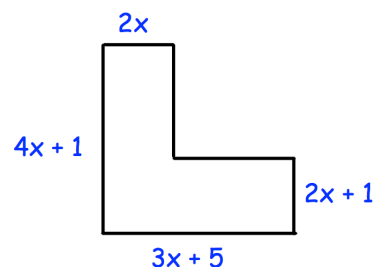
Question 1: Write down the perimeter of each shape below



Question 2: A square has a side length of $3x$.
Find an expression for the perimeter of the square.

Question 3: $6x + 7y + x - 8y = 7x - y$
Write down three other expressions that are equal to $7x - y$

Question 4: Find an expression for the perimeter of this shape



Answers



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Green band | G6

Expanding Brackets

Corbettmaths Practice Worksheet | www.corbettmaths.com

Name: _____

Level 2 Further Maths

Expanding Brackets



Corbettmaths

Ensure you have: Pencil or pen

Guidance

1. Read each question carefully before you begin answering it.
2. Check your answers seem right.
3. Always show your workings

Revision for this topic

www.corbettmaths.com/more/further-maths/



1. (a) Expand and simplify $2(3x + 1) + 4(9 - x)$

.....
(2)

- (b) Expand $w^4(w^2 + 3)$

.....
(2)

-
2. Expand and simplify $5(x - 2) - 2(4x - 3)$

.....
(2)

-
3. Expand and simplify $(m - 3)(2m + 3)$

.....
(3)

4. Expand and simplify $(3x + 5y)(7x - 2y)$

.....
(3)

5. Expand and simplify $(4x + 1)^2 - (4x - 1)$

.....
(3)

6. Expand and simplify $(6y - 5)(3y + 2) + (1 - y)(2 - y)$

.....
(3)

7. Expand and simplify $(2x + y)^2 - (2x - y)^2$

.....
(3)

8. Expand and simplify $(x^2 + 3x - 4)(3x - 4)$

.....
(3)

9. Expand and simplify $2xy(x + 2y)(3x - y)$

.....
(3)

10. $ax - 2(x + b) + 8 = 10(x + 2)$

$a = \dots\dots\dots$ $b = \dots\dots\dots$
(4)

11. $2a(3x - 1) + 3(ax + 7) \equiv 36x + b$

Find the values of a and b

$a = \dots\dots\dots$ $b = \dots\dots\dots$
(4)

12. (a) Expand $(y + p)(y - q)$

.....
(1)

(b) $y^2 + ay + b \equiv (y + p)(y - q)$

Write a and b in terms of p and q

$a =$

$b =$
(2)

13. Expand and simplify $(x + 4)(4x - 3) - 2(x - 5)^2$

.....
(3)

14. Simplify $(6x + 15)^2 - (5x - 10)^2 + 20x - 1$

.....
(4)

15. Expand and simplify $(4xy + 3xy^2 - 2y)(7x + x^2)$

.....
(4)

16. Expand and simplify $\frac{2}{x}(2x^3 + \frac{x^2}{2} + 3x)$

.....
(3)

Green band | G6

Expanding Brackets (Further)

Corbettmaths Practice Worksheet | www.corbettmaths.com

Examples



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Workout

Question 1: Expand the following brackets

- | | | | |
|------------------|-------------------|------------------|-----------------------|
| (a) $5(y + 3)$ | (b) $4(a + 2)$ | (c) $8(w + 10)$ | (d) $3(x - 7)$ |
| (e) $9(s - 1)$ | (f) $2(8 - t)$ | (g) $7(4 + h)$ | (h) $10(a + 2b + 3c)$ |
| (i) $4(3y + 2)$ | (j) $5(2p - 1)$ | (k) $3(7a + 2)$ | (l) $9(2x - 5)$ |
| (m) $5(4 + 3t)$ | (n) $7(9 - 2c)$ | (o) $8(3w + 1)$ | (p) $9(1 - 4p)$ |
| (q) $11(2k - 5)$ | (r) $20(6a + 5c)$ | (s) $3(15w - 7)$ | (t) $3(9 - 2a)$ |

Question 2: Expand the following brackets

- | | | | |
|-----------------|------------------|------------------|-------------------|
| (a) $-2(w + 5)$ | (b) $-3(c + 7)$ | (c) $-8(c + 7)$ | (d) $-10(y - 2)$ |
| (e) $-7(g - 3)$ | (f) $-4(2w + 3)$ | (g) $-9(3w - 5)$ | (h) $-9(5x - 1)$ |
| (i) $-5(6 - c)$ | (j) $-6(4 + 3m)$ | (k) $-2(1 + 9c)$ | (l) $-5(8a - 7w)$ |

Question 3: Expand the following brackets

- | | | | |
|------------------|------------------|------------------|-------------------|
| (a) $a(c + 2)$ | (b) $c(d - 3)$ | (c) $a(b + c)$ | (d) $w(8 - y)$ |
| (e) $c(5 + a)$ | (f) $w(a - 9)$ | (g) $y(s + t)$ | (h) $2a(c - 3)$ |
| (i) $5x(y + 8)$ | (j) $3a(2c + 9)$ | (k) $6g(2c - 1)$ | (l) $9k(2 + d)$ |
| (m) $5(2f + 9w)$ | (n) $3y(5p + 2)$ | (o) $2s(t + 1)$ | (p) $-4a(8x - 3)$ |

Question 4: Expand the following brackets

- | | | | |
|-----------------|------------------|------------------|-------------------|
| (a) $a(a + 2)$ | (b) $y(y - 5)$ | (c) $w(a + w)$ | (d) $c(9 - c)$ |
| (e) $p(2p + 5)$ | (f) $2w(3w - 1)$ | (g) $9y(2y + 3)$ | (h) $4c(2a + 5c)$ |

Expanding Brackets

Video 13 on www.corbettmaths.com

- (i) $2u(3 - u)$ (j) $m(m^2 + 3)$ (k) $y(y^2 - 7)$ (l) $g^2(g - 8)$
 (m) $2w(w^2 + 6)$ (n) $4a(2a^2 - 3)$ (o) $5c(3c^2 - a)$ (p) $8w(3w^2 + 3y)$
 (q) $x^2(x^2 + 4)$ (r) $3w^2(7 + 2w^2)$

Question 5: Expand and simplify

- (a) $5(y + 3) + 2(y + 7)$ (b) $6(2w + 5) + 9(w + 2)$ (c) $3(y - 2) + 4(2y + 5)$
 (d) $7(2g + 3) - 5(g + 2)$ (e) $6(x - 2) - 4(x - 8)$ (f) $2(3y - 8) - 5(2y - 1)$
 (g) $8(5 + 2m) + 3(5 - 3m)$ (h) $4(w + 7) - 2(2w + 1)$ (i) $9(1 + 2y) + 3(3 - y)$

Question 6: Expand and simplify

- (a) $w(w + 5) + w(w + 7)$ (b) $2g(4g + 3) + g(g - 7)$ (c) $n(n - 4) - n(5 - n)$
 (d) $2e(4e + 3) - 3e(e - 5)$ (e) $a(3 + c) + c(a + 2)$ (f) $m(a + 7) - a(4 - 3m)$
 (g) $8c(8 - 3a) + 3(4 - c)$ (h) $5y(3y + z) - 2y(4y - 3z)$ (i) $4c(3c - c^2) - 2c^2(4 - 5c)$

Apply

Question 1: Can you spot any mistakes in the questions below.

Expand $3(2y - 1)$

$$6y - 1$$

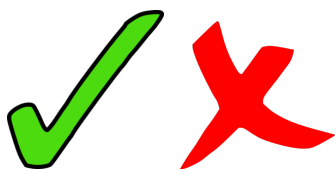
Multiply out $x(x + 3)$

$$2x + 3x = 5x$$

Expand and simplify $6(w + 3) - 2(w - 5)$

$$\begin{aligned} &6w + 18 - 2w - 10 \\ &= 4w + 8 \end{aligned}$$

Answers



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